

Overview and Economic Evaluation of Single Use Technology

Jeff Johnson
Biotech Design LLC
Biotechdesign@outlook.com
BiotechdesignLLL.com

Nov 14, 2023

Agenda

- Single Use Overview and Trends
- Net Present Cost (NPC) vs. Cost of Goods (COGs)
- Simple SS vs. SU Equipment Economic Evaluation
- Traditional SS vs. SU / Hybrid Facility of the Future Economics
 - Key Inputs
 - Model Results
- Conclusions
- Questions and Discussion

Single Use Prior to ~2010

- Limited Scale and Applications
 - Single Use Bags up to ~200L
 - Single Use Assemblies made in-house and autoclaved
- SS before SU on capital projects
- No SUBs, SUMs or DACs in licensed GMP Commercial Facilities
- Custom in house autoclaved machines
- Focus on Extractables and Process Specific Leachables
- Very limited SU Sensors

Single Use Current State

- Widespread use of 3000L Bags
- Widespread use of DACs
- SUMs up to 3000L+
- SUBs routinely used up 2000L and now available up to 6000L
- Ready to use gamma irradiated assemblies made to order
- SU Sensors available, but limited sizes and lower accuracy than SS
- Extractables & Leachables assessment well understood
 - BPOG E&L Protocol
- SU common on capital projects
- Procedures and execution approach not as consistent as SS
- Perception that SU vs. SS varies between areas
 - SU does not require CIP/SIP
 - SU requires set-up every batch, CIP/SIP of SS is fully automated
- Facility Integration is custom on each project

Facility Integration – Current State



Integration of SU Components,
Tubing Management, and
Automation Hardware & Software
Needs Industry Wide Solutions



Single Use in the Future

- SU on all projects, including Small Molecule
- SU and SU Hybrid Facilities common at all scales
- Standard designs from stock inventory
 - Standardization allows multiple suppliers, reducing supply chain risk
- Bags, SUMs and SUBs $\geq 5000\text{L}$
 - SU Tubing $>1''$ readily available at low cost from multiple suppliers
- SU continuous processing with SUBs $\leq 1000\text{L}$ in perfusion
 - SU Tubing $<1/4''$
- SU Sensors for all applications and sizes
- Standard facility integration solutions
- Perception that SU is faster, simpler and lower cost than SS

Net Present Cost vs. Cost of Goods

- **Output primarily presented in Net Present Cost (NPC)**
 - NPC is a discounted cash flow analysis – money spent now is worth more than in the future
 - Discount rate used is company proprietary. ~10% is a reasonable estimate currently
 - NPC recognizes the total expense associated with the facility including construction, capital and operating costs for a brownfield facility assuming typically 10-20 years of operation
 - 15 years in the equipment assessment
 - When comparing options, the lowest NPC will be the most favorable financial option
 - NPC is based on cash flow as opposed to P&L or COGs timing
 - COGs accounts for capital as straight line depreciation over many years
 - Depreciation varies by asset type – Facility vs. Equipment. Often estimated by using 15 – 20 years
 - COGs will show less sensitivity to capital than NPC, as capital is front loaded in NPC calculation
 - NPC is used to evaluate business decisions comparing multiple alternatives with different cash flow profiles
 - NPC is used when revenue is not a variable/consideration as it only considers cash spend
 - Net Present Value is used when revenue is a variable.
 - For manufacturing option analysis, can use NPC as manufacturing will provide all goods required to meet sales need – i.e. revenue is not variable

- **Simplified example**

- Assume a \$100MM capital investment over 3 years (20%, 60% and 20% spending curve)
- Assume 1000kg/year production starting in year 4
- Assume \$50MM in operating expense, starting in first year of operation
- Assume 3% inflation
- Assume 10% discount rate (Varies company by company, 10% is reasonable)
- Assume 20 year straight line depreciation

[illegible]

Simple SU vs. SS Economic Evaluation

Net Present Cost (NPC) Analysis

Spreadsheet developed to compare cost of Single Use (SU) vs. Stainless Steel

Lower upfront capital costs for SU

Operating cost savings (no CIP/SIP, etc.) vs. higher disposable costs

Capital and operating costs calculated for 15 years

NPC calculated from capital and cost estimates

Example

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$60
Validation Labor Rate \$/hr	\$120
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$50.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$2,000
Fully installed capital - SU	\$80,000
Fully installed capital - SS	\$900,000
Depreciation, years	15
Annual Lots	20
	NPV
Total before taxes	(\$634,576)
Total after taxes	(\$573,592)
Negative NPV favors single use.	

NPC Analysis of SS vs. SU Equipment

- **Results are highly dependent on Operating Costs and Run Rate**
 - Capital can be important (such as for bioreactors), but typically not as important
 - Operating costs driven by cost of disposables
 - SS operating costs (CIP/SIP, etc) typically not significant
 - Run Rate coupled with operating costs usually drives results
 - A low run rate (say 20 batches/year) and low disposables favors SU due to low capital
 - SU Bioreactors are almost always favored due to low run rate of ~20/year
 - SU Fermentors more limited to clinical production, as commercial operation can approach 100 batches / year
 - A high run rate and high disposables cost favors SS
 - SU Chromo and TFF more limited to clinical as wetted path costs >\$10k and ~100 batches per year commercial

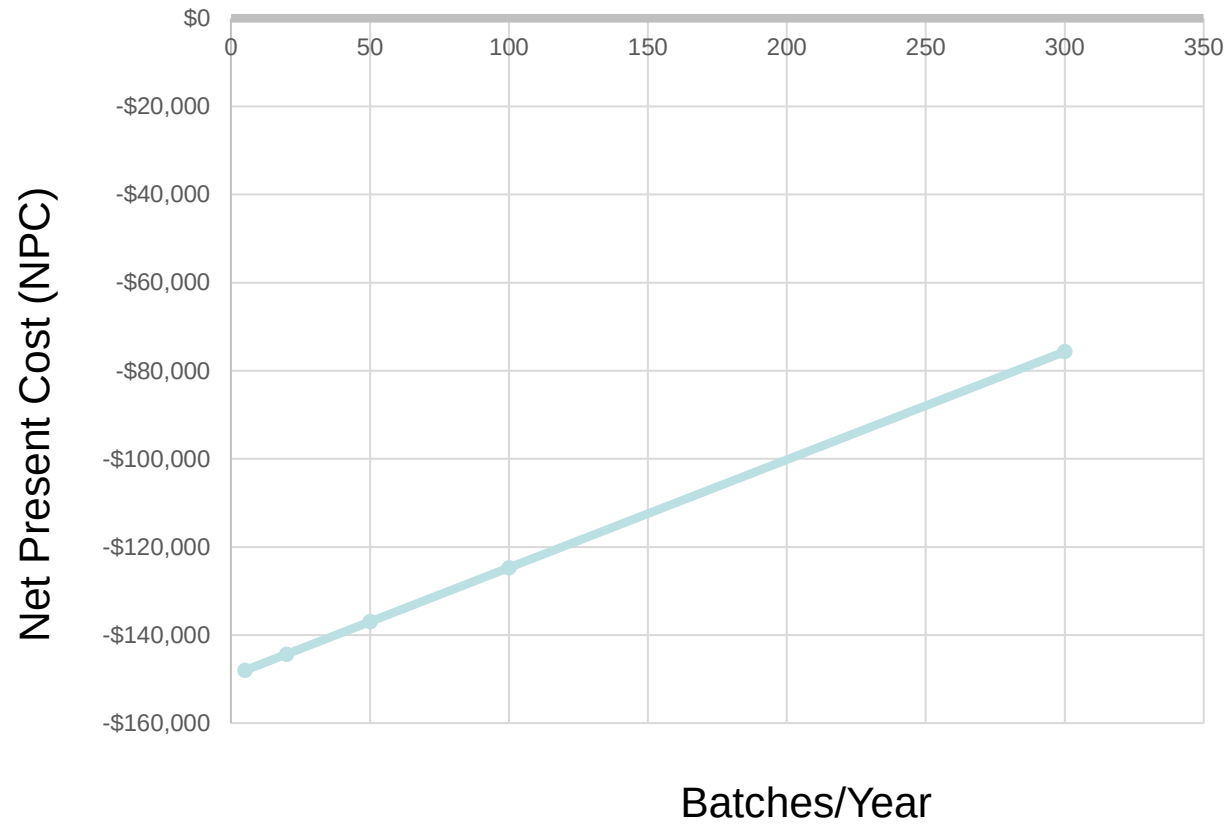
NPC Analysis of SS vs. SU (con't)

- **Rough “Rules of Thumb” of SS vs. SU equipment (up to limit of scale)**
 - SU Bags w/o agitators almost always favorable to SS
 - SU Mixers favorable at low run rates (<50/yr), SS favorable at higher run rates (>100 yr)
 - Buffer prep tanks SS in SU facility (hybrid design)
 - SU Bioreactors almost always favorable due to long batch times if scale is adequate
 - Higher titer greatly helps as scale limitation is not as great an impact
 - SU Fermenters are more limited due to high run rate, limited scale and higher costs
 - Can be highly preferred in R&D and Clinical applications
 - SU Chromo and TFF are limited due to high disposables costs and low flows
 - Hybrid designs with SS skids that are self CIP, requires no CIP skids, fully SU in and out
- Case by case evaluation should be made during Concept / BOD of new facility design

200L Buffer Hold SS vs SU Bioreactor Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$60
Validation Labor Rate \$/hr	\$120
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$50.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$250
Fully installed capital - SU	\$1,000
Fully installed capital - SS	\$100,000
Depreciation, years	15
Annual Lots	5
	NPV
Total before taxes	(\$175,061)
Total after taxes	(\$148,009)
Negative NPV favors single use.	

200L Buffer Hold SS vs. SU Bioreactor – Run Rate vs. NPC



Favors SS



Favors SU

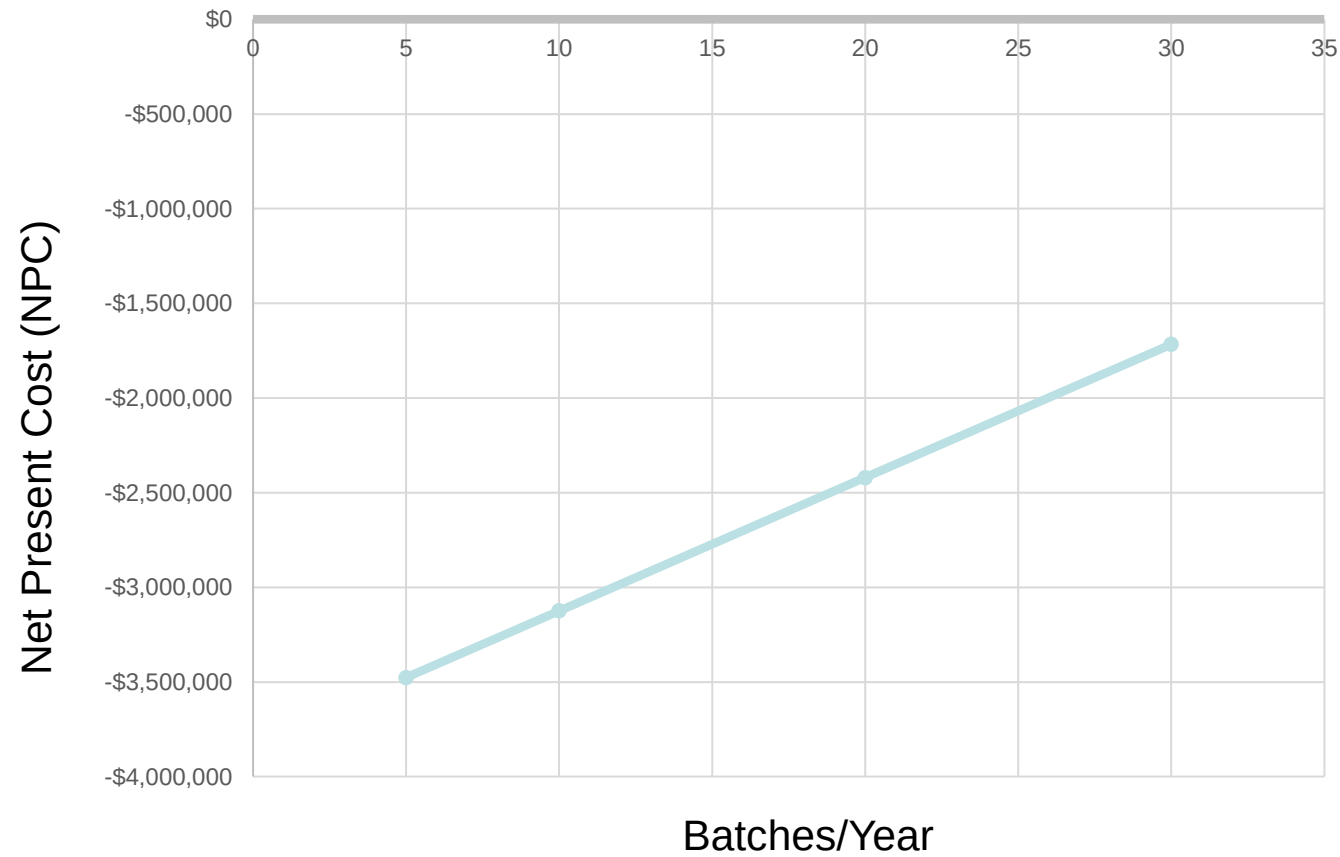


2000L SS vs SU Bioreactor Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$50
Validation Labor Rate \$/hr	\$100
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$20.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$1,000
Fully installed capital - SU	\$1,000,000
Fully installed capital - SS	\$5,000,000
Depreciation, years	15
Annual Lots	25
	NPV
Total before taxes	-\$4,829,546
Total after taxes	-\$4,535,540

Negative NPV favors single use.

2000L SS vs. SU Bioreactor – Run Rate vs. NPC



Favors SS



Favors SU

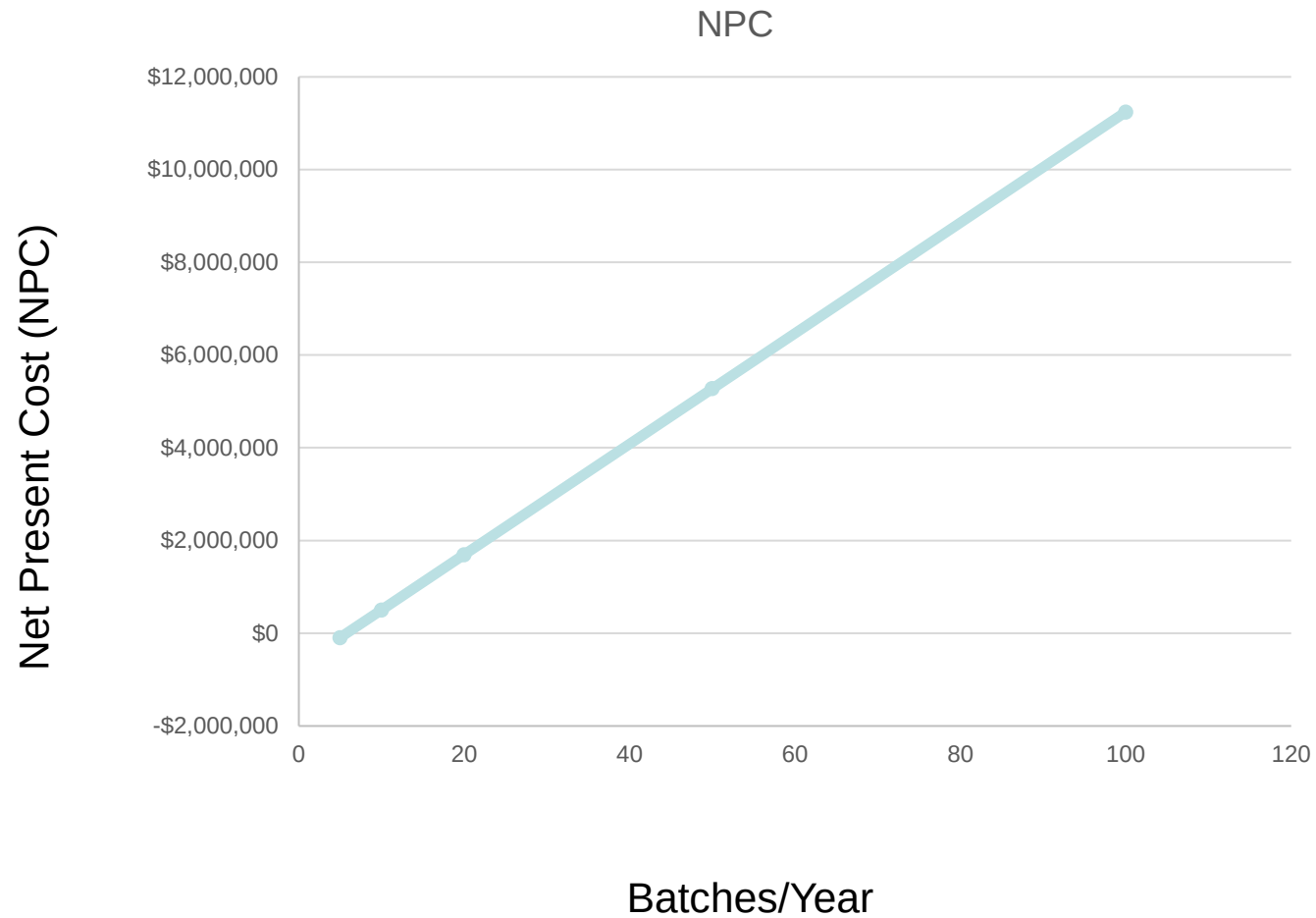


SS vs SU Chromatography Skid Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$50
Validation Labor Rate \$/hr	\$100
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$20.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$15,000
Fully installed capital - SU	\$1,000,000
Fully installed capital - SS	\$1,000,000
Depreciation, years	15
Annual Lots	5
	NPV
Total before taxes	-\$74,039
Total after taxes	-\$94,776

Negative NPV favors single use.

SS vs. SU Chromatography Skid – Run Rate vs. NPC



Favors SS



Favors SU



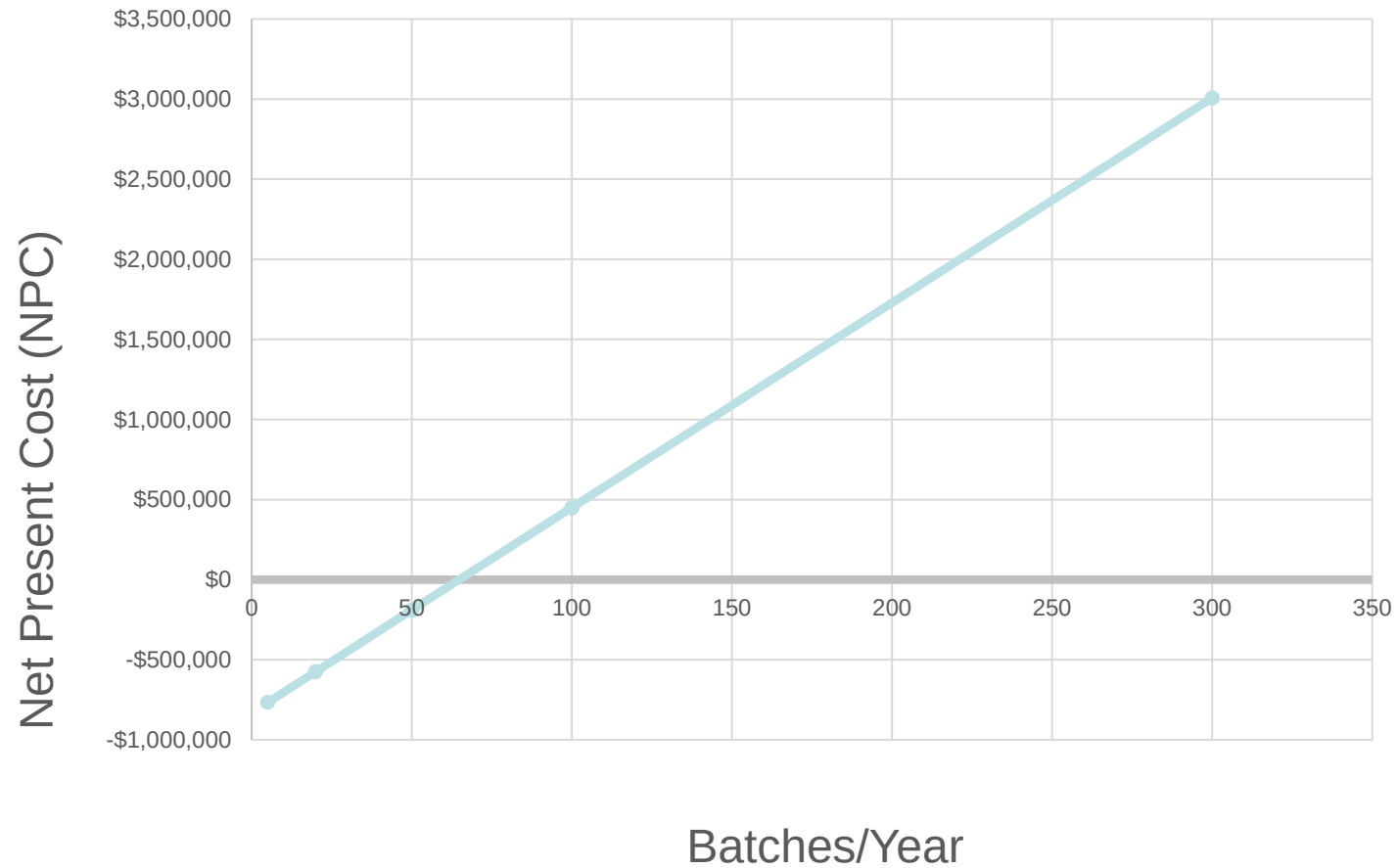
3000L SS vs SU Buffer Hold Tank Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$60
Validation Labor Rate \$/hr	\$120
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$50.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$2,000
Fully installed capital - SU	\$80,000
Fully installed capital - SS	\$900,000
Depreciation, years	15
Annual Lots	20
	NPV
Total before taxes	(\$634,576)
Total after taxes	(\$573,592)
Negative NPV favors single use.	

Buffer Hold Tank with no mixer

- No platform for SU
- Platform required for SS

3000L SS vs SU Buffer Hold Tank Data Input



Favors SS



Favors SU



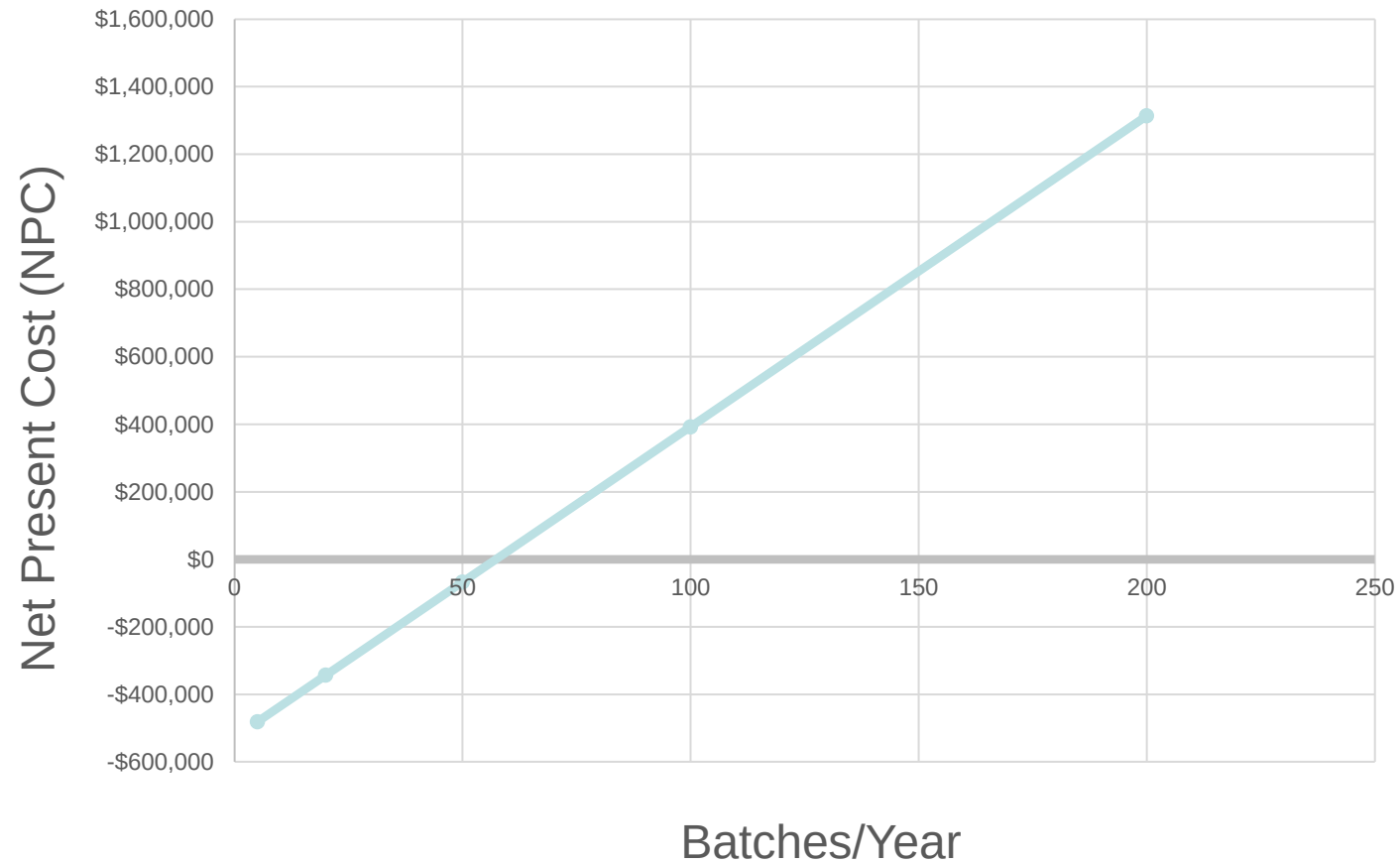
Note: Break even point is sensitive to price of SU Disposable cost

1000L SS vs SU Mixer Process Tank Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$60
Validation Labor Rate \$/hr	\$120
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$50.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$1,500
Fully installed capital - SU	\$250,000
Fully installed capital - SS	\$750,000
Depreciation, years	15
Annual Lots	20
	NPV
Total before taxes	(\$379,019)
Total after taxes	(\$343,417)
Negative NPV favors single use.	

Process Tank with no platform

1000L SS vs SU Mixer Process Tank Data Input



Favors SS



Favors SU



Note: Break even point is sensitive to price of SU Disposable cost

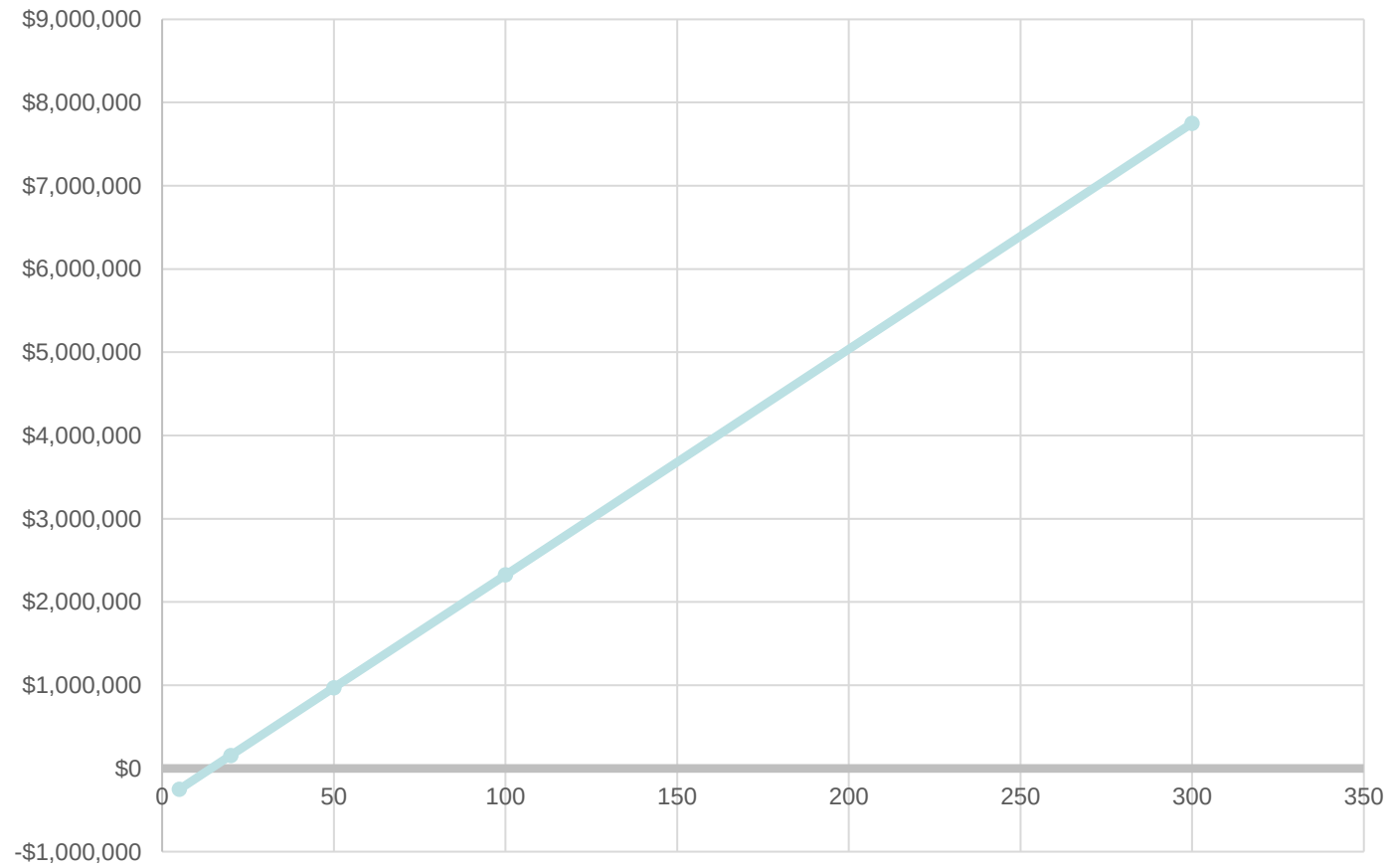
2000L SS vs SU Mixer Buffer Prep Tank Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$60
Validation Labor Rate \$/hr	\$120
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$50.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$4,000
Fully installed capital - SU	\$900,000
Fully installed capital - SS	\$1,250,000
Depreciation, years	15
Annual Lots	20
	NPV
Total before taxes	\$230,645
Total after taxes	\$156,375
Negative NPV favors single use.	

Buffer / Media Prep Tank

- Platform for both SS and SU
- Drum Dumping Lift for both SS and SU
- Assumes CIP skid is already installed and available
 - Required for transfer lines, etc
- Traditional SU Mixer
- Capital and SU cost estimates from IPS

2000L SS vs. SU Mixer Buffer Prep Tank – Run Rate vs. NPC



Favors SS



Favors SU



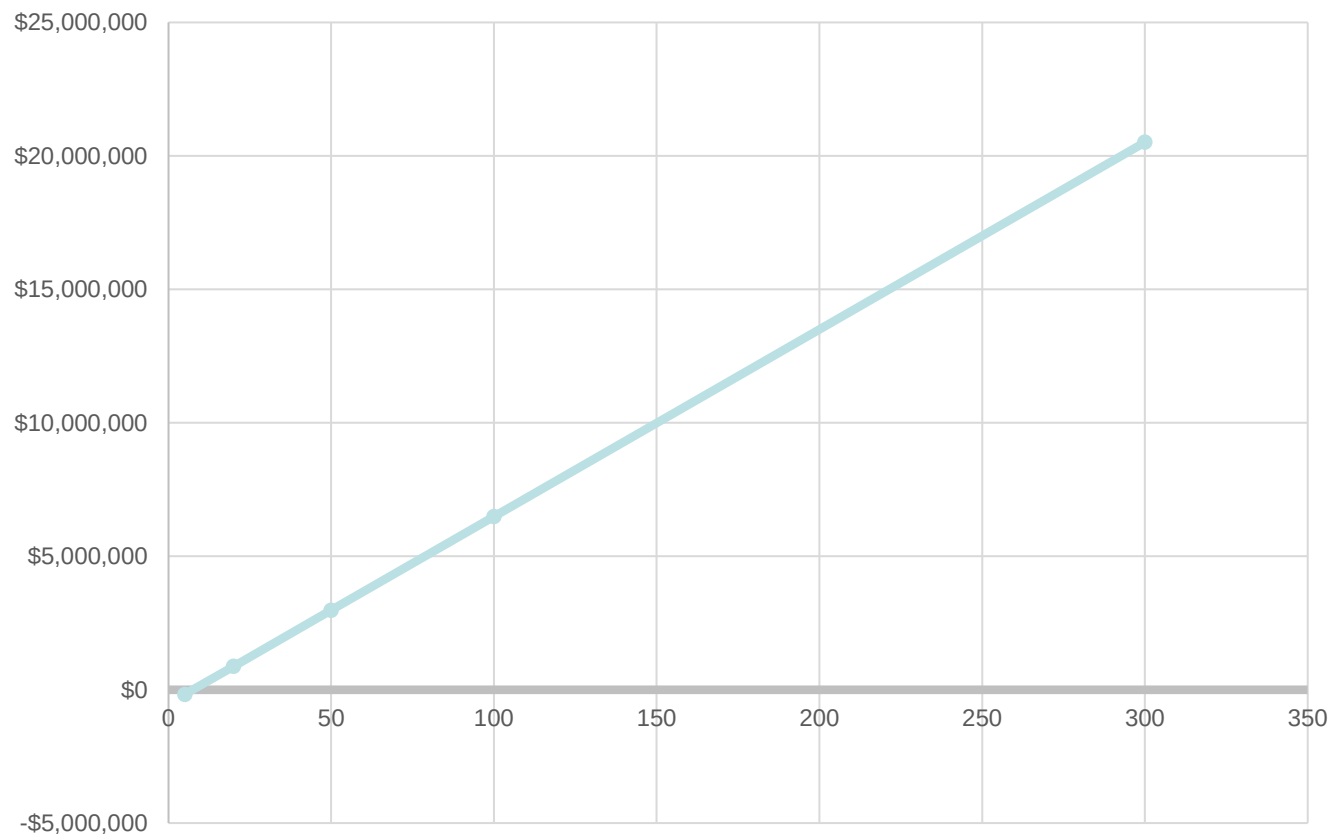
5000L SS vs SU Mixer Buffer Prep Tank Data Input

Factor	
Escalation %	2.0%
Escalation For Single Use, %	3.0%
Operator Labor Rate \$/hr	\$60
Validation Labor Rate \$/hr	\$120
High Purity Water, \$/liter	\$0.06
Liters WFI per CIP	1500
Clean Steam, \$/kLb	\$40.00
Clean steam per SIP Lb	150.00
CIP Detergent Cost, \$/Liter	\$2.00
CIP Detergent Liters/Run	5
Liquid waste disposal cost, \$/m3	\$5.00
Single use disposal cost \$/item	\$50.00
Labor Hours per CIP	2
Labor Hours per SIP	1
Labor hours per bag set up/break down	1
Annual Maintenance Cost, % TIC	2.0%
Discount Rate, %	10.0%
Tax Rate On Profits, %	20.0%
Qualification hrs per CIP	320
Qualification hrs per SIP	320
SU Costs per lot	\$10,000
Fully installed capital - SU	\$1,000,000
Fully installed capital - SS	\$1,500,000
Depreciation, years	15
Annual Lots	20
	NPV
Total before taxes	\$1,143,781
Total after taxes	\$874,823
Negative NPV favors single use.	

Buffer / Media Prep Tank

- Platform for both SS and SU
- Drum Dumping Lift for both SS and SU
- Assumes CIP skid is already installed and available
 - Required for transfer lines, etc
- TFS Impulse SU Mixer
- Capital and SU cost estimates from IPS

5000L SS vs. SU Mixer Buffer Prep Tank – Run Rate vs. NPC



Favors SS



Favors SU



Discussion and Optional Topics

Q&A for single piece of equipment economics

Optional Topics if Time Allows:

- Overall Facility Economic Evaluation
- Seed Train Strategy example for Fed Batch 15kL Example

Overall Facility Economic Evaluation

- **Savings of SS vs SU is limited on equipment basis**
 - Savings on the order of ~\$10(s)MM
- **Savings on a “Modular Plug and Play” Facility can be >\$100M**
 - Eliminate CIP and SIP
 - Eliminate extensive WFI and Clean Steam piping
 - In many cases, SS equipment is favorable, but keep to “Modular Plug and Play” design
 - Design “self cleaning” to eliminate CIP skids and interconnecting piping
 - Make hybrid to simplify any CIP requirements (SS main piping with SU Feeds and Product lines)
 - All fermenter or large bioreactor feeds are single use (perhaps with Lynx valve connections)
 - TFF is SS for main feed and retentate line only – all feeds and permeate lines SU
 - Chromo is self cleaning with bags of cleaning chemicals and WFI
 - Goal is to attach bag of cleaning chemical and water, with no CIP skid required
 - If not possible, then use small portable CIP skid

Current Fed Batch Processes and Model

- **Existing traditional Fed Batch Mab process**
 - 6 bioreactors with 3 seed trains
 - Depth Filtration Harvest
 - ProtA Chromo
 - VI
 - AEX
 - CEX
 - VF
 - TFF
- **2kL to 15kL SS Facilities**
- **2kL to 5kL SU Facilities**

Case Study: Cost of Goods Analysis at Full Capacity

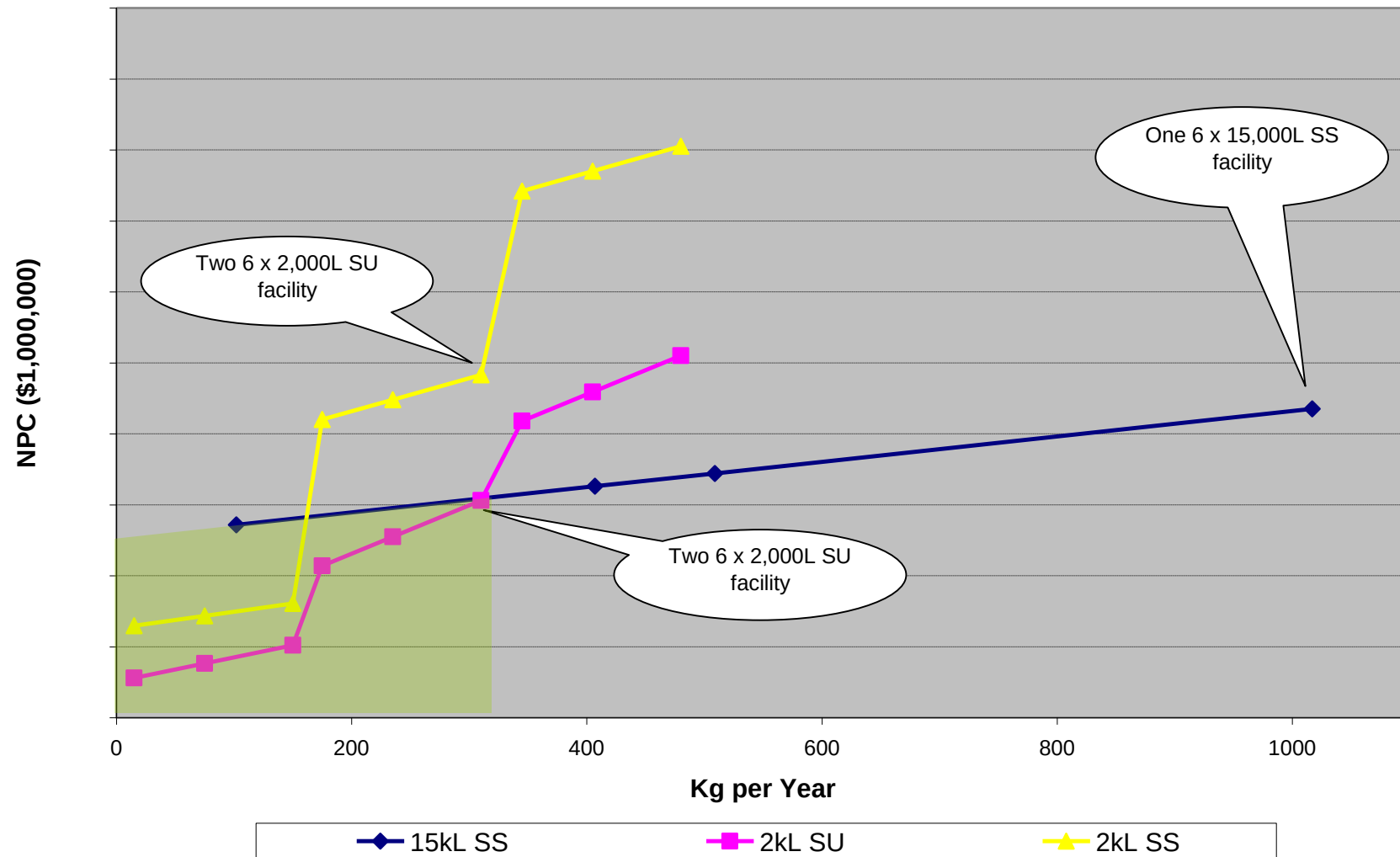
Platform	Bioreactor Size (L)	Batches / Year	Utilization (%)	Titer (g/l)	KG / Year	\$ / Gram
SS	15,000	120	100	3	3,500	\$52
SS	2000	120	100	3	470	\$129
SU	2000	120	94	3	470, - - -	\$102

Case Study: Cost of Goods Analysis at ~470kg/yr Capacity

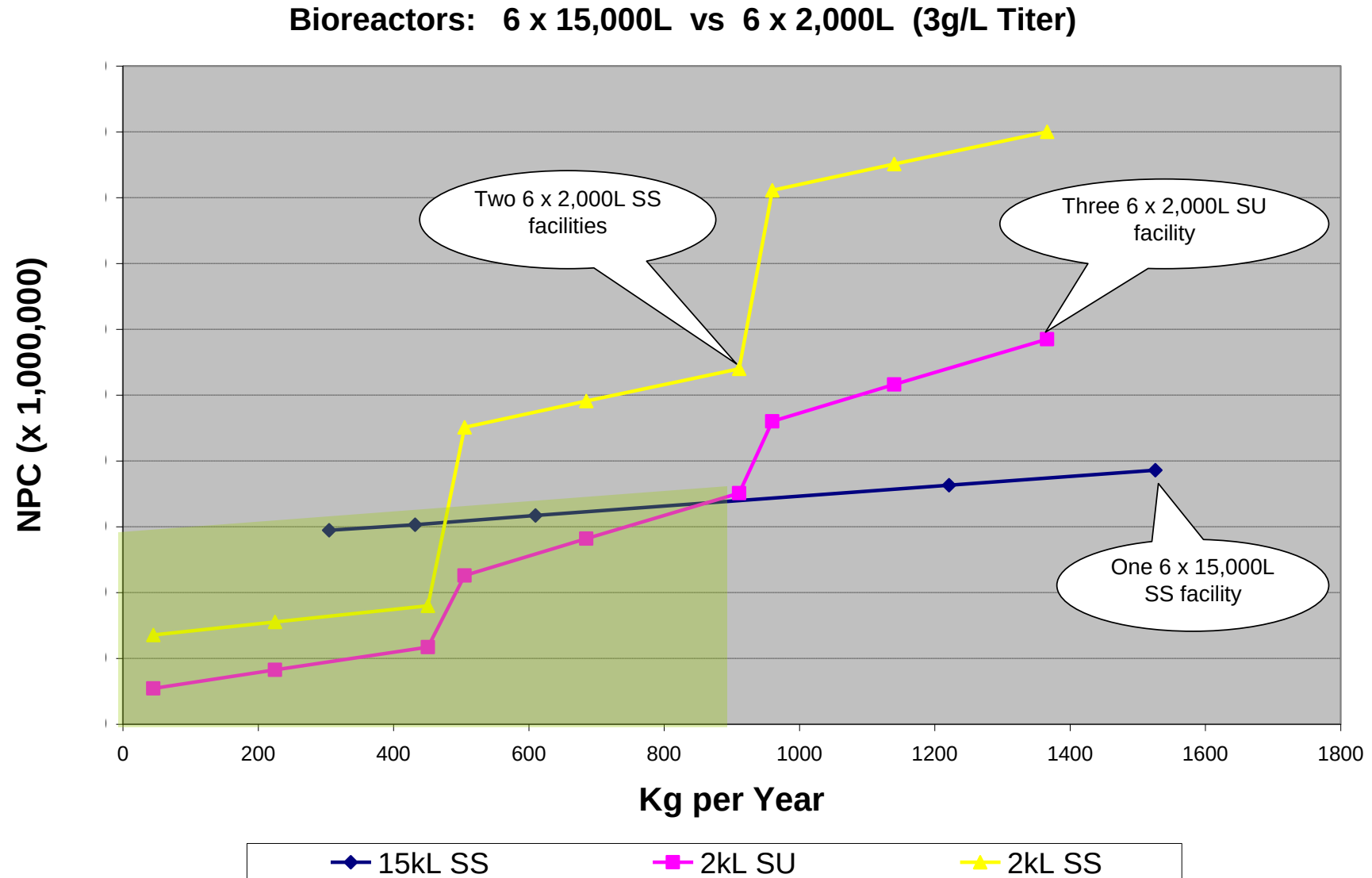
Platform	Bioreactor Size (L)	Batches / Year	Utilization (%)	Titer (g/l)	KG / Year	\$ / Gram
SS	15,000	120	100	3	3,500	\$52
SS	15,000	17	14	3	470	\$174
SS	2000	120	100	3	470	\$129
SU	2000	120	94	3	470	\$102

Full SS vs. Full SUT Facility Economics – 1g/l Titer

Bioreactors: 6 x 15,000L vs 6 x 2000L (1g/L Titer)

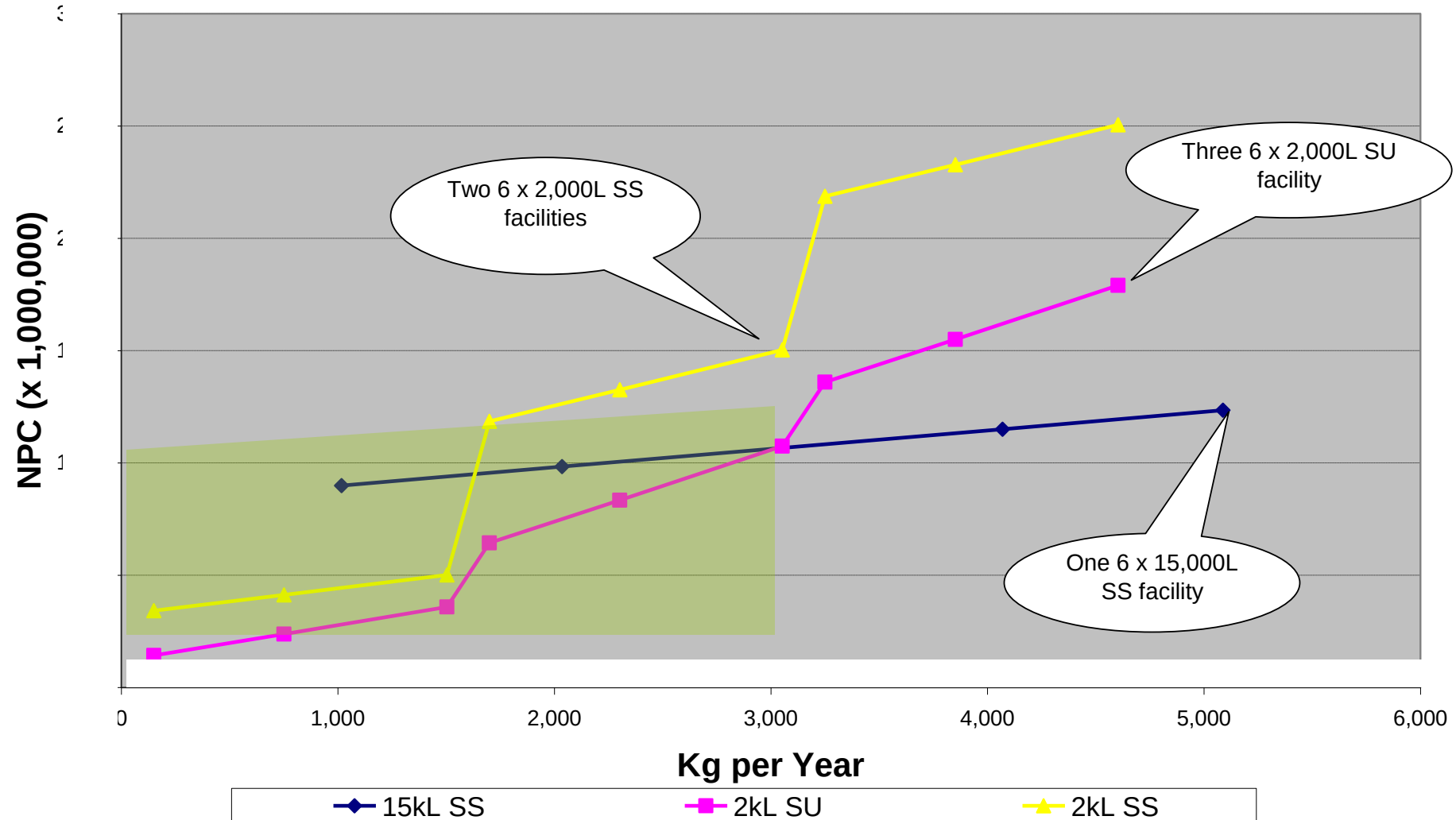


Full SS vs. Full SUT Facility Economics – 3g/l Titer

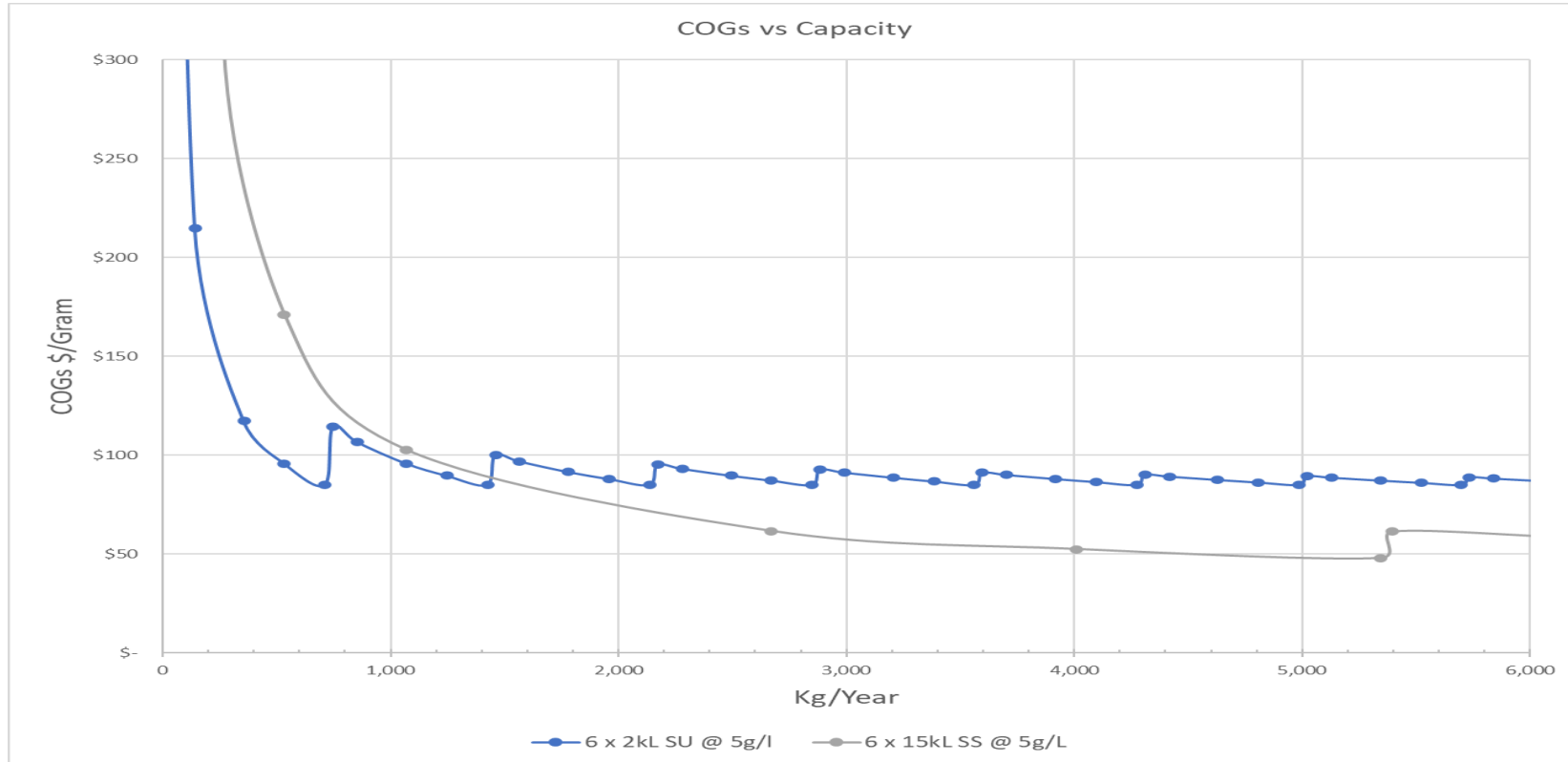


Full SS vs. Full SUT Facility Economics – 10g/l Titer

Bioreactors: 6 x 15,000L vs. 6 x 2,000L (10 g/L Titer)



COGs vs Capacity – SS and SU at 5g/l



Note: SU crosses SS after 2nd Facility vs. 3rd Facility in NPC analysis
COGs discounts the impact of capital vs. NPC

Next Generation Processes and Model

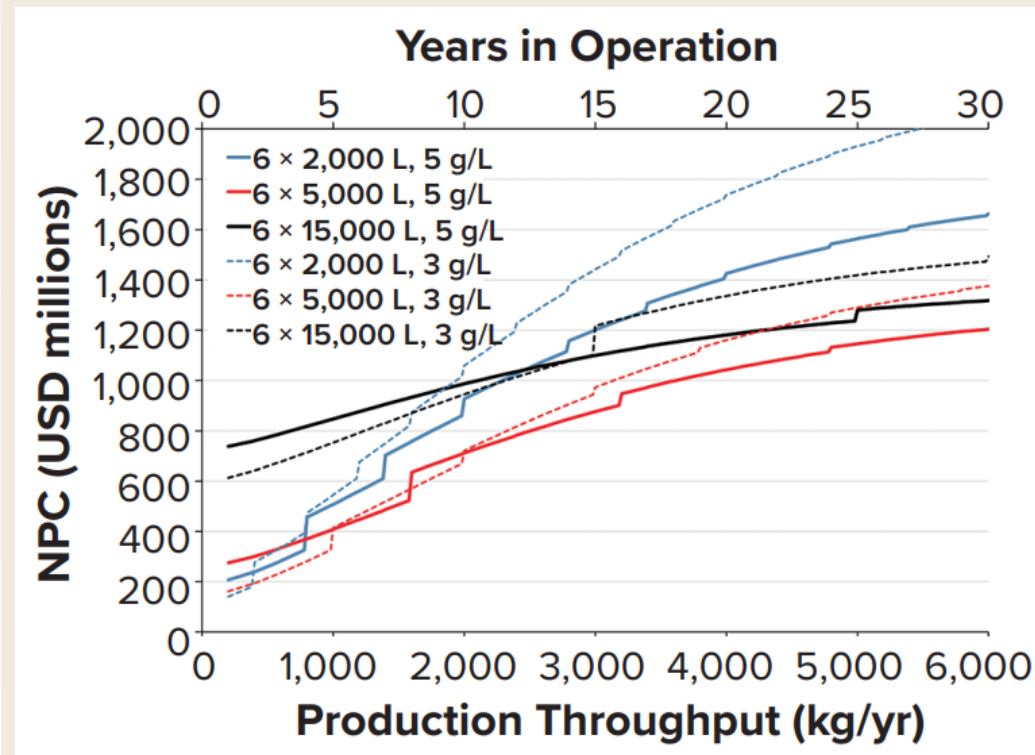
- **Next Generation Process Economic Evaluation**
 - Enhanced model to include facility ramp rate
 - Intensified Batch USP, Perfusion USP and Continuous DSP

Impact of Facility Ramp-up

- NPC graphs shown so far assume that the build for all facilities needed to reach a target capacity are started in same year, and all turn on at full capacity
 - Not an accurate depiction
 - Requires assumptions on product demand and launch schedules
 - How much protein is required for each product
 - What is the ramp rate of each product
 - How often are new products launched?
 - Requires a more complicated model to handle ramp rate
 - Following model results assume a consistent ramp up in demand of 200kg/yr
 - Result is a curvature in the results due to discounting of future capacity
 - Favors Single Use as lower capital initially and higher operating costs in the future are deferred
- Ramp Rate can also be incorporated into COGs analysis
 - Also favors SU, but not as much as NPC as initial capital spend is underweighted due to accounting for capital via straight-line depreciation

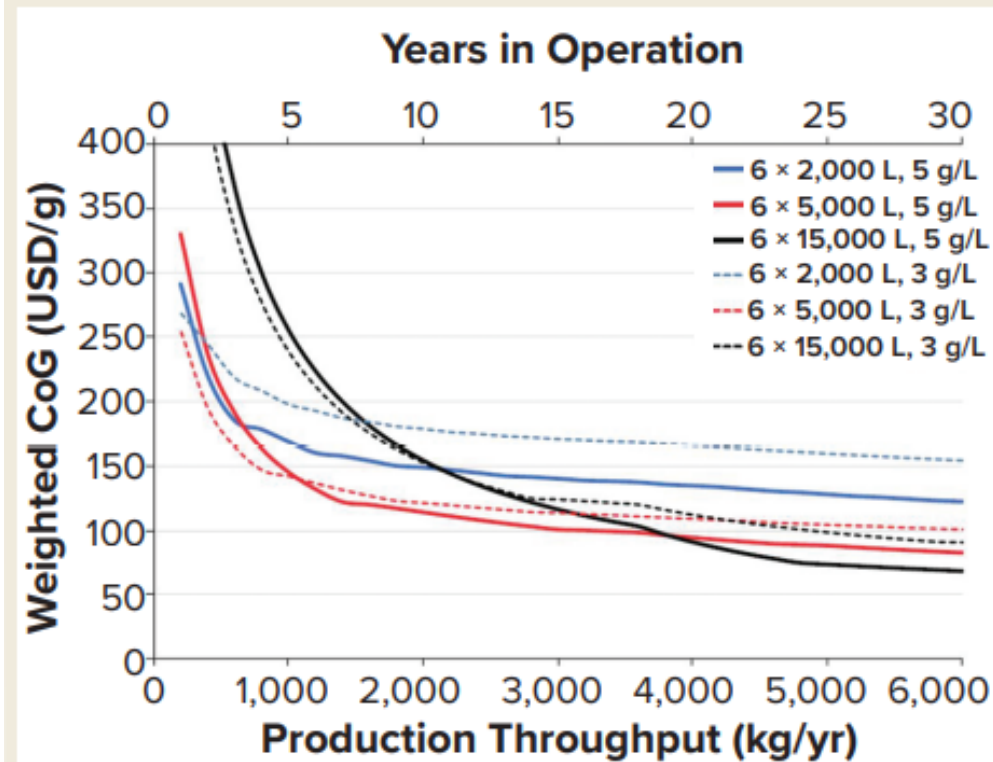
Production Ramp Rate impact on NPC

Figure 1: Perspective of total facility ownership costs in terms of net present cost (NPC); discounted operating costs are captured in the primary slopes of the lines. Steps in the lines represent discounted costs of a new facility required to support production demand. Data are at a 200-kg/yr/yr production ramp.



Production Ramp Rate impact on COGs

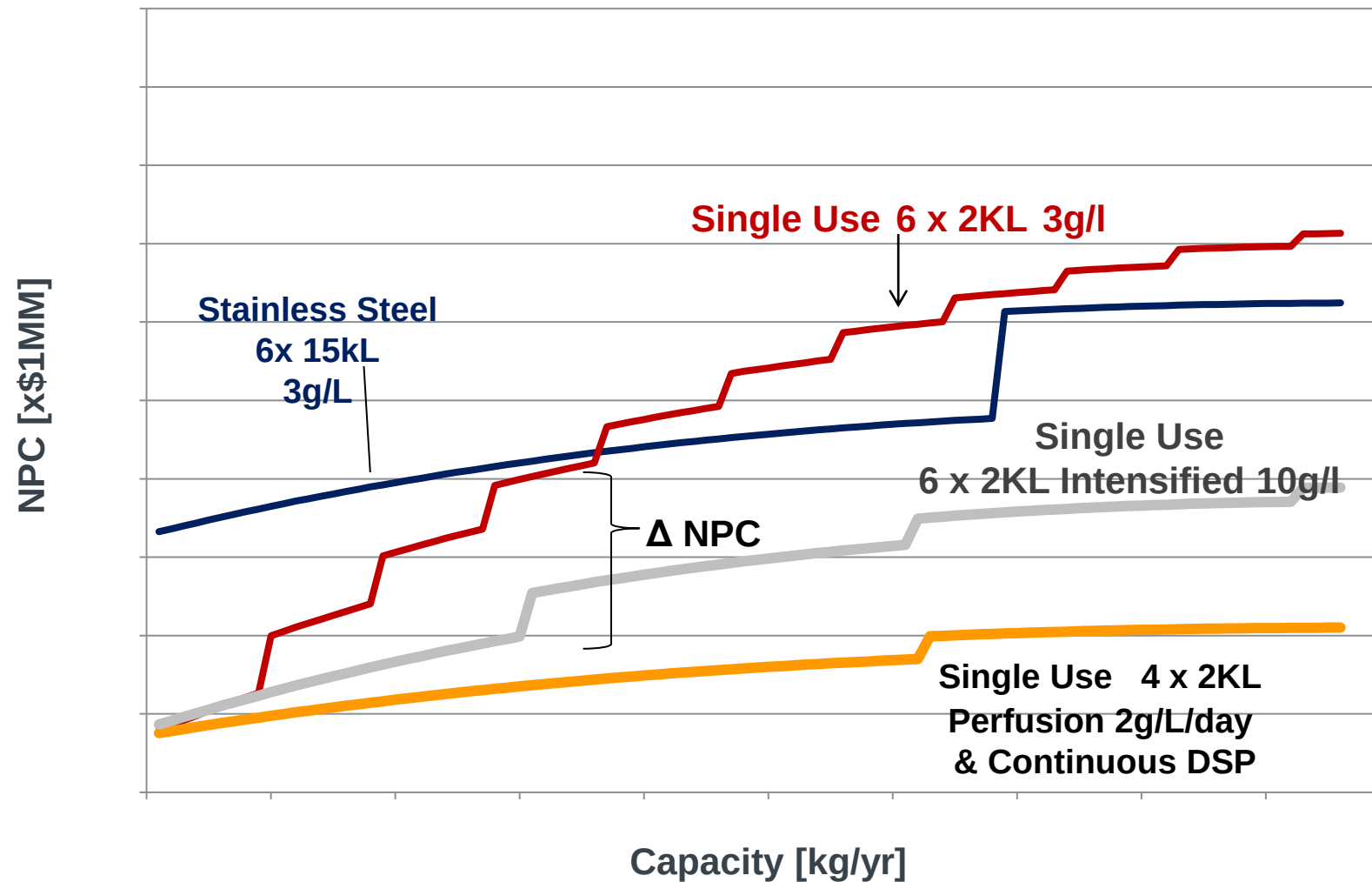
Figure 2: Weighted cost of goods (CoG) for greenfield facility builds; CoG in year 1 are weighted by CoG of previous years to highlight actual net present CoG at the given year and production throughput. Graph displays weighted CoG with a production ramp rate of 200 kg/yr/yr.



Source: Updating the Economics of Biologics Manufacturing with 5000-L Single-Use Bioreactors A Paradigm Shift. Mark Thomas et. al. BioProcess International 18(10)si Oct 2020

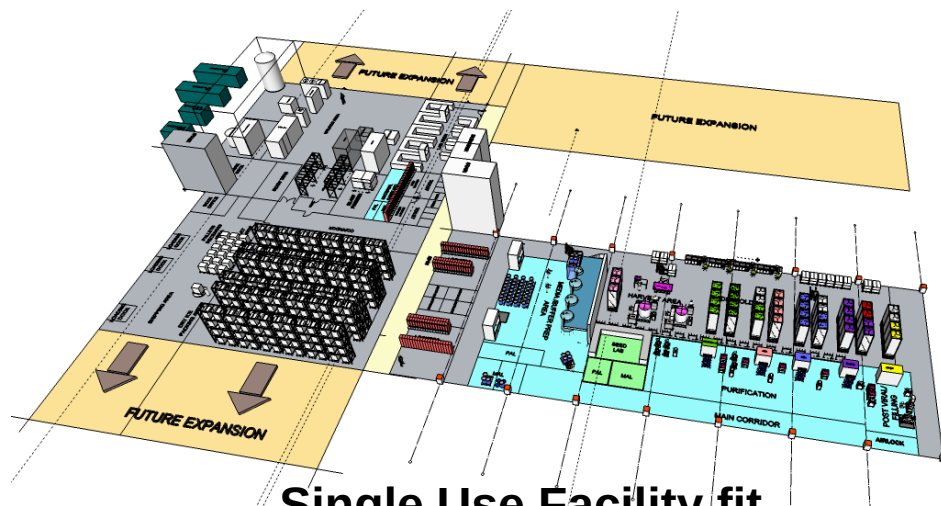
Yes! SU Facilities offer Significant Savings over SS!

Significant NPC savings from SU Intensified & Perfusion Processes



SU Facility Concepts

- SU based Platform at up to 2,000 L scale
 - Intensified fed batch 3-5x titer increase
 - Perfusion with consistent harvest titer
- Modular facility design & construction
 - Scale out rather than scaleup
 - SU allows simpler designs & faster construction schedules
 - Quick to duplicate modules in multiple locations
- Consistent Process Scale allows faster Technology transfer
 - 2,000L SU Bioreactor scale used for all work – clinical, commercialization, launch

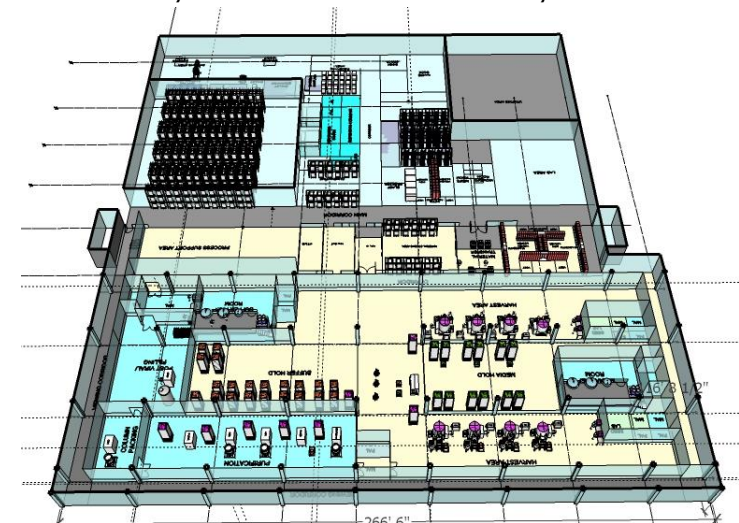


Single Use Facility fit



Public

r 2 x 2,000L Perfusion SUBs with Continuous DSP



Single Use Facility fit

for 6x 2,000L SUBs & Batch DSP

Seed Train Strategy Economic Evaluation

- Traditional SS seed train strategy may not be optimal for SU facilities
 - Traditional SS approach based off high capital costs and short seed cycle time
 - Traditional SS approach:
 - One seed train feeds two production reactors since seed cycles times were $< 50\%$ of production
 - Current trend towards N-1 perfusion may not fit this model
 - Lower capital cost of SU Bioreactors allow for different approach:
 - One seed train per bioreactor – allows for N-1 perfusion times equal to production time
 - Higher turndown ratios in newer SUBs ($\geq 5:1$) allows for two seed steps in one SUB
 - Initial inoculation at low media fill level, then top off with media for next stage
 - Economic evaluation can help decide on optimal strategy

Seed Train Strategy Economic Evaluation

- Traditional SS seed train strategy may not be optimal for SU facilities
 - Traditional SS approach based off high capital costs and short seed cycle time
 - Traditional SS approach:
 - One seed train feeds two production reactors since seed cycles times were $< 50\%$ of production
 - Current trend towards N-1 perfusion may not fit this model
 - Lower capital cost of SU Bioreactors allow for different approach:
 - One seed train per bioreactor – allows for N-1 perfusion times equal to production time
 - Higher turndown ratios in newer SUBs ($\geq 5:1$) allows for two seed steps in one SUB
 - Initial inoculation at low media fill level, then top off with media for next stage
 - Economic evaluation can help decide on optimal strategy

Seed Train Strategy Economic Evaluation Case Study

- Four seed train options evaluated for a new 4 x 15kL SS Bioreactor Facility
- Baseline traditional SS Approach:
 - Custom SS Bioreactors for each split, one seed feeds two production reactors:
 - N-4 = 60L SS x 2
 - N-3 = 235L SS x 2
 - N-2 = 950L SS x 2
 - N-1 = 3750 SS x 2
 - N = 18,500L SS x 4
- Option 1: SUB up to 2000L (N-1 SS)
- Option 2: SUB up to 5000L (N-1 SU)
- Option 3: SUB up to 5000L, one seed train per bioreactor, two stages in each SUB
 - N-4 = NA – use N-3
 - N-3 = 500L SU DynaDrive 10:1 Turndown x 4
 - N-2 = NA – use N-1
 - N-1 = 5000L SU DynaDrive 10:1 Turndown x 4

Seed Train Strategy Economic Evaluation Results

	NPC	Annual \$
Baseline	\$25MM	0
Option 1	\$24MM	\$1.4MM
Option 2	\$26MM	\$2.7MM
Option 3	\$20MM	\$1.7MM

- Option 3 selected for Basis of Design, even though unconventional:
- Lowest NPC and reasonable annual operating costs
- Dedicated seed train per production bioreactor gives operational advantages:
 - Reduced schedule complexity
 - Allows for N-1 perfusion cycle times to approach production cycle time
- Conclusion: Economic evaluation is an important tool to help in Facility Conceptual and Basis of Design

Conclusions

- Economic Evaluation of SS vs. SU Equipment is a valuable tool
- Hybrid Approaches may yield optimal outcomes
- Economic Modeling of SU/Hybrid Factory of the Future shows great promise
 - Single-use fed-batch 6x2,000L at $\geq 3\text{g/L}$ titer is cost effective over stainless steel
 - Lower initial capital and deferring operating expense until demand requires
- Industry collaboration to achieve full potential of Single Use Technology
 - Standardization
 - Facility Integration / Tubing Management